

Algebra-Based Physics-1: Midterm 1

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1 Unit 0: Estimation, Unit Analysis, Vectors, and Kinematics I

- Which of the following represents a typical flow rate for a bathroom faucet?
 - A: 0.8 liters per minute
 - A: 8.0 liters per minute
 - A: 80.0 liters per minute
 - A: 800 liters per minute
- A train leaves Union Station in Denver, CO, heading for Chicago, IL. If the destination is 1000 miles to the East, how long before the train arrives?
 - A: 0.2 hours
 - B: 2.0 hours
 - C: 20.0 hours
 - D: 200.0 hours
- What is 0.001 m s^{-2} in cm min^{-2} ?
 - A: 3.6 km hr^{-1}
 - B: 36.0 km hr^{-1}
 - C: 360 km hr^{-1}
 - D: 3600 km hr^{-1}
- Suppose a ship accelerates from 0 km hr^{-1} to 10 km hr^{-1} in 60 seconds. What is the acceleration?
 - A: $60 \text{ km hr}^{-1} \text{ s}^{-1}$
 - B: $6 \text{ km hr}^{-1} \text{ s}^{-1}$
 - C: $1/6 \text{ km hr}^{-1} \text{ s}^{-1}$
 - D: $1/60 \text{ km hr}^{-1} \text{ s}^{-1}$
- Estimate the area of the North Quad of Whittier College (the open space outside the SLC):
 - A: 5000 m^2
 - B: 5000 cm^2
 - C: 500 m^2
 - D: 500 cm^2

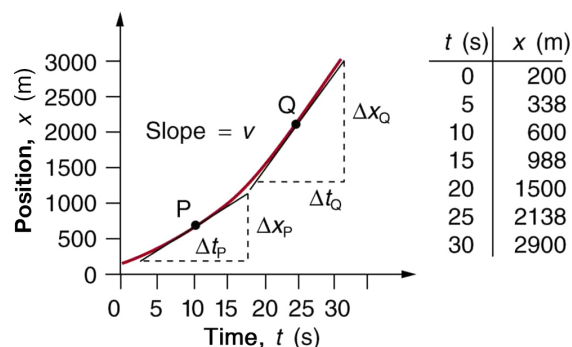


Figure 1: A system moves with non-constant velocity.

- (a) Let $\vec{v} = -2\hat{i} - 2\hat{j}$, and $\vec{w} = 2\hat{i} - 2\hat{j}$. Draw each in a 2D coordinate system below. (b) What is $\vec{v} + \vec{w}$? (c) What is $\vec{v} - \vec{w}$? (d) What is $\vec{v} \cdot \vec{w}$?

2 Unit 1: Kinematics II and III

- Suppose a cyclist has a velocity of 15 m s^{-1} at $t = 0$. If the acceleration is 3 m s^{-2} , (a) what is the velocity at $t = 4$ seconds? (b) What is the displacement of the cyclist at $t = 4$ seconds? (c) Are the average and instantaneous velocities different at these times?
- Consider the motion data shown in Fig. 1. What is the acceleration of the system?

3. A system moves according to $x(t) = 12t - 3t^2$ m. (a) What is the average velocity between these times? (b) What is the acceleration?

4. **Design problem.** Design an experiment in which a baseball is thrown, and the range must be 60 meters. Choose the launch angle and initial velocity, and show that the range is 60 meters. Provide the time of flight as well. Finally, verify your results with the appropriate PhET simulation.

5. **Design problem.** In our lab, we used a pendulum to measure g , the gravitational constant. Use the following simulation to repeat that process: <https://phet.colorado.edu/en/simulations/pendulum-lab>. Recall that the formula relating period T to g is $T = 2\pi\sqrt{L/g}$, where L is the pendulum length. Show your work in the form of a (handwritten) graph and relevant calculations.

3 Unit 2: Forces I and II

1. A 14,500 kg F/A-18 Super Hornet lands on an aircraft carrier, moving at 230 km hr^{-1} . A tow cable grabs the aircraft and pulls it to a stop in 330 meters. (a) What is the average acceleration? (b) What force does the tow cable exert to stop the jet?

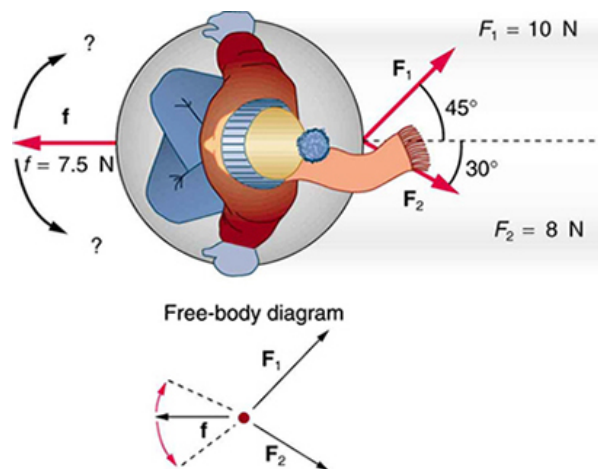


Figure 2: Two people pull on a third person on a sled, on an icy surface.

2. Two children pull a third child on a snow saucer sled exerting forces $\vec{F}_1$ and $\vec{F}_2$ as shown in Fig. 2. Find the acceleration of the 36 kg sled and child system.
3. Now consider the sled and child in the previous exercise, sliding down a hill with the friction coefficient for steel on ice ($\mu = 0.02$). (a) If the hill has a 10 degree incline, what is the acceleration? (b) How far will the child and sled travel in 20 seconds?
4. Consider four springs connected *in parallel* to an object of mass m . Each spring has a spring constant k , and each spring is attached to the floor and the object. (a) Draw a free-body diagram. (b) Derive an expression for the displacement of the springs. (c) Show that, in the limit that $k \rightarrow \infty$, the displacement goes to zero. This is a basic model for vehicle suspension. (d) Verify your results with the Hooke's Law PhET simulation. Show that two springs *in parallel* with the same k-value act as one spring with twice the k-value.

4 Unit 3: Forces III and IV

1. What is the terminal velocity of a 60 kg skydiver with area $A = 0.25 \text{ m}^2$, and drag coefficient $C = 0.5$? Use the standard density of air: $\rho = 1.2 \text{ kg m}^{-3}$. (b) What is the terminal velocity if she opens the parachute, increasing the cross-sectional area by a factor of 100?
2. Calculate the change in length of the upper leg bone (the femur) when a 55.0 kg woman supports 45.0 kg of her mass on it, assuming the bone to be equivalent to a uniform rod that is 30.0 cm long and 2.00 cm in radius. Young's Modulus for bone is $9 \times 10^9 \text{ N m}^{-2}$.
3. Find the mass of a picture hanging from a steel nail, given that the nail bends only 4.5 microns (10^{-6} m). The shear modulus is $80 \times 10^9 \text{ N m}^{-2}$, and the diameter of the nail is 3.00 mm.
4. **Scaling Problem:** Consider how we finished the exercise on the bone and Young's modulus. Scale this exercise to apply to a steel beam supporting a building. The beam is 20 meters long, has a radius of 10 cm, and supports $\approx 10^4 \text{ kg}$. What is the change in length?